

Influence of Running Shoe Longitudinal Bending Stiffness on Running Economy and Performance in Trained and National Level Runners

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ABSTRACT

RODRIGO-CARRANZA, V., W. HOOBKAMER, J. J. SALINERO, S. RODRÍGUEZ-BARBERO, J. M. GONZÁLEZ-RAVÉ, and F. GONZÁLEZ-MOHÍNO. Influence of Running Shoe Longitudinal Bending Stiffness on Running Economy and Performance in Trained and National Level Runners. *Med. Sci. Sports Exerc.*, Vol. 55, No. 12, pp. 2290–2298, 2023. **Introduction/purpose:** Previous results about shoe longitudinal bending stiffness (LBS) and running economy (RE) show high variability. This study aimed to assess the effects of shoes with increased LBS on RE and performance in trained and national runners. **Methods:** Twenty-eight male runners were divided into two groups according to their 10-km performance times (trained, 38–45 min and national runners, <34 min). Subjects ran 2 × 3 min (at 9 and 13 km·h⁻¹ for trained, and 13 and 17 km·h⁻¹ for national runners) with an experimental shoe with carbon fiber plate to increase the LBS (Increased LBS) and a control shoe (without carbon fiber plate). We measured energy cost of running (W·kg⁻¹) and spatiotemporal parameters in visit one and participants performed a 3000 m time trial (TT) in two successive visits. **Results:** Increased LBS improved RE in the trained group at slow (11.41 ± 0.93 W·kg⁻¹ vs 11.86 ± 0.93 W·kg⁻¹) and fast velocity (15.89 ± 1.24 W·kg⁻¹ vs 16.39 ± 1.24 W·kg⁻¹) and only at the fast velocity in the national group (20.35 ± 1.45 W·kg⁻¹ vs 20.78 ± 1.18 W·kg⁻¹). The improvements in RE were accompanied by different changes in biomechanical variables between groups. There were a similar improvement in the 3000 m TT test in Increased LBS for trained (639 ± 59 vs 644 ± 61 s in control shoes) and national runners (569 ± 21 vs 574 ± 21 s in control shoes) with more constant pace in increased LBS compared with control shoes in both groups. **Conclusions:** Increasing shoe LBS improved RE at slow and fast velocities in trained runners and only at fast velocity in national runners. However, the 3000 m TT test improved similarly in both levels of runners with increased LBS. The improvements in RE are accompanied by small modifications in running kinematics that could explain the difference between the different levels of runners. **Key Words:** FOOTWEAR, SUPER SHOES, RUNNING ABILITY, RUNNING PERFORMANCE, CARBON FIBER PLATE

Advanced footwear technology (AFT) such as the Nike Vaporfly 4% or similar shoe models are characterized by the presence of a curved carbon fiber plate in the midsole of the shoe to increase the longitudinal bending stiffness (LBS), a thicker and more compliant resilient midsole foam and with more rocker shape (1). Combined, these shoe characteristics have been shown to improve running economy (RE) (2–5) and performance (6–8) with small kinematic changes such as increased stride length and increased contact time (9). Running economy is a key factor on long running performance, however,

when running at intensities above the metabolic steady state (e.g., a 3000 m test at near maximum oxygen uptake intensity (10)), RE is not the main indicators of performance so performance tests or the evaluation of race pace are needed to evaluate the effect of footwear in short-distance events (11).

Although for commercial shoes that combine a carbon fiber plate with compliant and resilient midsole foam, group average improvements in RE of up to 4% have been reported (compared with traditional racing shoes), reported RE improvements from increased LBS *by itself* are less pronounced [for review see Ortega et al. (9) and Rodrigo-Carranza et al. (12)]. This might be because many previous studies increased the LBS of shoes with external insoles or flat carbon fiber plates and not with a curved plate embedded in the midsole foam (13–17).

For AFT shoes, previous studies have reported greater RE improvements in highly trained/national runners (sub-32:00 men and 35:30 women for 10 km) (2–4) than in recreational/trained runners (~45:00 for 10 km) (5,18–20). A recent meta-analysis showed a RE improvement in world-class and amateur runners with the greatest variability of results in the

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world-class runners (21). Although this variability can be partly due to comparing RE values across multiple laboratory visits and unstandardized control shoes, these observations suggest that runners of different caliber might not all benefit in the same way from AFT shoes (22). It is well known that not all runners respond biomechanically in the same way to an increase in LBS, particularly with differences in metatarsophalangeal and ankle joints kinematics (13,14,23), possibly related to biomechanical and anthropometric characteristics (2), or plantar-flexor muscle strength (13) while running at sub-maximal intensities. This suggests that the effect of LBS on RE could be specific to the caliber and speed of the runner (19,24).

This raises the question whether shoes with increased LBS affect RE and distance performance differently between groups of national and trained runners. Therefore, the aim of the current study was to examine the influence of shoes with increased LBS on RE, spatiotemporal parameters and short-distance performance between trained and national runners. We hypothesized that national runners would show larger improvements in RE and short distance performance with similar spatiotemporal modifications, when running in shoes with increased LBS, compared with trained runners.

METHODS

Participants. Twenty-eight male runners (mean $\pm$ SD, 28.1 ± 6.4 yr; 64.8 ± 5.6 kg and 174.9 ± 5.0 cm) participated in this study. The inclusion criteria were the following: 1) participation in running training for at least 3 d $\cdot$ wk $^{-1}$ during the previous 6 months without injury, 2) ability to run 10-km in less than 45 min, 3) fitting a men's US 9 to 11 shoe size. Exclusion criterion was inability to run at 13 km $\cdot$ h $^{-1}$ (for the trained runner group) or 17 km $\cdot$ h $^{-1}$ (for the national runner group) while below the onset of blood lactate accumulation of 4 mmol $\cdot$ L $^{-1}$ and with respiratory exchange ratio (RER) values below 1.0 during all the tests (2). One subject of the trained group was excluded from the final analysis due to a blood lactate reading greater than 4 mmol $\cdot$ L $^{-1}$.

The participants were divided into two groups according to the classification made by McKay et al., (25) and to their 10-km performance level: national runners group ($n = 14$; mean $\pm$ SD: 27.5 ± 7.7 yr; 62.3 ± 4.0 kg and 175.4 ± 5.4 cm; $31:51 \pm 0:53$ (range, 29:25–33:40) [mm:ss] in 10 km) with a time below 34 min in 10-km and the trained runner group ($n = 13$; mean $\pm$ SD: 28.4 ± 4.7 yr; 67.4 ± 6.0 kg and 174.3 ± 4.8 cm; $38:51 \pm 0:51$ (range: 38:02–42:23) [mm:ss] in 10-km) with a time between 38 and 45 min in 10-km. All subjects had previously been AFT users.

Twenty-seven participants completed the first session involving familiarization and RE assessment during treadmill running. Eight participants did not complete at least one of the 3,000 m time trials (TT) (second and third sessions) due to injuries ($n = 3$) or personal reasons ($n = 5$), four of which were from the group of national runners and 4 from the group of trained runners and were eliminated from the analysis of the 3000 m TT test.

Before the study, all participants were informed about the testing protocols, possible risks involved and were invited to provide written informed consent. The study was performed in accordance with the principles of the Declaration of Helsinki, and the experimental protocols were approved by the local ethics committee (UNNE-2021-010).

Shoes characteristics. An experimental shoe model with a curved full carbon fiber plate inserted in the middle of the midsole to increase LBS and a control shoe model without carbon fiber plate were specifically manufactured for this study. Both shoe models were identical in the composition and characteristics (last, upper, midsole foam (ethylene-vinyl acetate [EVA]), midsole hardness, color, and shape), with the aim to control a possible placebo effect (4). The manufacturer labeled the shoes with a red or black mark according to the two shoe conditions (control or experimental), and this was used by the research team to keep the study subjects blinded. Only the senior researcher and first author knew which condition each mark corresponded to, so that when collecting data, the other researchers, and runners themselves did not know which condition



FIGURE 1—Shoe conditions. (A) Top: control, bottom: increased LBS. (B) Left: control, Right: increased LBS.

TABLE 1. Shoe characteristics, comfort and experience.

Characteristics	Control	Experimental
Carbon plate	No	Full curved carbon fiber plate
LBS (N·mm ⁻¹)	19.12 ± 0.43	35.21 ± 0.59
Mass (g)	240.0 ± 4.0	248.3 ± 5.5
Heel Stack (mm)	39.1 ± 0.2	39.13 ± 0.23
Forefoot Stack (mm)	31.2 ± 0.3	31.1 ± 0.1
Heel-Toe offset (mm)	8.0 ± 0.1	8.1 ± 0.1

Shoes characteristics (of the sizes 9–11 US) represents mean ± SD.

was tested (Fig. 1). Shoe characteristics are presented in Table 1. The average mass of the shoes tested was 240.0 ± 4.0 g and 248.3 ± 5.5 g for control and experimental condition respectively. Even though, additional shoe mass is known to deteriorate RE and running performance by about 1% per 100 g (26,27), we have shown that changes of up to 50 g do not affect performance at high running intensity (28). All athletes had their own individual experimental and control pair of shoes of their size to control the possible effect of shoe wear on LBS.

Experimental setup and protocol. The effect of increased LBS on RE and performance was evaluated using a randomized crossover experimental design. All participants refrained from intense exercise in the 48 h preceding the test sessions: a first session in the laboratory (controlled environmental conditions [550 m altitude, 20–22°C, 35–40% relative humidity]) and two sessions at an outdoor athletics track, all separated by 7 d (controlled environmental conditions: 15–17°C, 25–30% relative humidity). Participants were asked to avoid extenuated exercise, caffeine and alcohol intake within 24 h of their testing visits and refrained from eating and drinking for 6 h before testing.

During the first session, RE and spatiotemporal parameters were assessed for two shoes models. The sequence of the shoes was randomized for each participant using a random number generator.

First, anthropometrical variables were measured. Height was measured to the nearest 0.1 cm with a portable stadiometer (Seca, Bonn, Germany) and body mass was measured to the nearest 0.1 kg with a portable balance (Seca, Bonn, Germany) while barefoot and wearing light shorts. Then, the researchers fitted the participant with the correct shoe size of both conditions, whereas the participant was not allowed to manipulate the footwear at any time. Before the RE testing trials, all participants completed 10 min of running as a warm-up in their own shoes at a self-selected pace slower than $9 \text{ km} \cdot \text{h}^{-1}$ for the trained runners and $13 \text{ km} \cdot \text{h}^{-1}$ for the national runners. At the end of the warm-up, the runners performed 2 efforts of 30 s at the fastest velocity of the test ($13 \text{ km} \cdot \text{h}^{-1}$ for the trained group and $17 \text{ km} \cdot \text{h}^{-1}$ for the national group).

Participants then completed 2×3 min trials at 9 and $13 \text{ km} \cdot \text{h}^{-1}$ for the trained runners group and 13 and $17 \text{ km} \cdot \text{h}^{-1}$ for the group of national runners with each type of shoe model (total $2 \times [2 \times 3 \text{ min}]$). We selected 3 min trials because steady-state $\dot{V}\text{O}_2$ can be reached within 2–3 min at each running velocity (29). The oxygen consumption between minute 2 and minute 3 of all participants differed $<0.5\%$, thus it was possible to confirm that the subjects were in metabolic steady state. The velocities selected were those that they were able to run below the lactate threshold to ensure $\dot{V}\text{O}_2$ steady-state mea-

surements. This was also confirmed by blood lactate measures (values below $4 \text{ mmol} \cdot \text{L}^{-1}$) and respiratory exchange ratio (values below 1.0). All running trials were performed on a motorized treadmill (HP Cosmos Pulsar, HP Cosmos Sports & Medical GMBH, Nussdorf-Traunstein, Germany) with a vertical stiffness value around 300 kN/m (30). The treadmill slope was set at 1% to more accurately reflect the energetic cost of outdoor running (31). During the RE trials, respiratory variables were measured using a gas analyzer (CPX Ultima Series MedGraphics, St. Paul, MN), which was calibrated before each session (CO_2 4.10%; O_2 15.92%). The gas analyzer was connected to a neoprene facemask (with a silicone adapter) that covers the subject's mouth and nose. $\dot{V}\text{O}_2$ values collected during the last 1 min of each 3 min trial was used to calculate RE in $\text{w} \cdot \text{kg}^{-1}$ (32–34). Blood samples were collected from the fingertip at the first minute after finishing each run trial for blood lactate determination (Lactate Scout, SensLab GmbH, Germany). At the end of the trial of each shoe condition, participants rested 10 min before the following trial with the other condition.

The main spatiotemporal parameters of the gait cycle (contact time, step frequency, step length and flight time, vertical oscillation) were measured for every step during treadmill running using a Stryd® Power Meter device, sampling at 1000 Hz. The device uses a triaxial accelerometer with adequate reliability and excellent validity of these measures evaluated have been reported by García-Pinillos et al., (35) and was placed on the right shoelaces of all subjects. The Stryd® device was paired with a Polar® watch (Vantage M GPS system; Polar, Kempele, Finland) and the information was analyzed using the Stryd Power Center program available on the web. Data were recorded and the second minute of each RE trial was averaged for subsequent analyses.

Finally, during the following two visits (separated by 7 d), the participants performed a 3000 m time trial (TT) in each shoe condition in a randomized order at an outdoor athletics track (400 m) to evaluate the influence of AFT on short distance running performance. Pacing during the 3000 m TT test was measured at 200 m and then every 400 m lap to compare the effect of pacing on TT running performance. The participants received verbal information about laps completed and number of laps to go, but no other feedback or encouragement was provided during the test. That is, participants were kept blind to time during each of the 3000 m TT test. Heart rate (HR) was measured continuously using a HR monitor with a paired watch (Polar Vantage V GPS system). Average HR for each test was calculated to provide an index of effort.

At the end of all experimental sessions, it was explained to the subjects that 1 d they had run in a shoe with a carbon fiber plate to increase the LBS and the other day in a shoe without carbon fiber plate (in a randomized order), and participants were asked whether they knew what shoes they had been tested in on each day. Only two runners noted differences between the shoes they had used each day and correctly identified the shoes they had run in each day, whereas most runners noted no difference between days.

Statistical analysis. Statistical analyses were carried out using IBM SPSS Statistics for Macintosh, Version 28.0 (IBM

Corp., Armonk, NY). Data were screened for normality of distribution and homogeneity of variance using a Shapiro–Wilk normality test. All variables were assessed with a three-way ANOVA with repeated measures in two factors (group–shoe–velocity). The sphericity assumption was checked with Mauchly’s test followed by the Greenhouse–Geisser adjustment where required. During 3000 TT test, the pacing was evaluated using a repeated measures ANOVA to compare the speed of the first lap with the rest of the laps in each shoe condition.

When differences were identified, pairwise comparisons with LSD adjustment were used. Results are expressed as mean $\pm$ standard deviation. Effect sizes (ES) were measured using partial eta squared in the ANOVAs and values of 0.01, 0.06, and above 0.15 were considered as small, medium and large, respectively (36) and Cohen’s *d* was used to compare between experimental conditions in each group with values of 0.2, 0.5, and above 0.8 were considered as small, medium, and large, respectively (36). Pearson’s correlation coefficients were used to investigate the relationship between RE in the control shoe and the change in RE with the experimental condition and between individual RE differences and 3000 TT tests performance differences (percentage change between shoe conditions). The significance level was set at $P < 0.05$ for all analyses.

RESULTS

During the RE assessment session, blood lactate at the conclusion of the final trial at 13 or 17 km·h⁻¹ was 1.56 ± 0.58 mmol·L⁻¹ (1.52 ± 0.45 and 1.61 ± 0.66 for trained and competitive, respectively) and was below 4 mmol·L⁻¹ for all subjects included in the final analysis. RER remained below 1.0 across all trials for all subjects (0.91 ± 0.06 in the final minute of the fastest velocity). Moreover, these velocities chosen for the group of trained and national runners resulted in similar relative intensities (RER = 0.92 ± 0.05 vs 0.91 ± 0.05 for trained and national runners respectively; $P = 0.69$). There was no indication of any substantial $\dot{V}O_2$ and $\dot{V}CO_2$ slow component, as

oxygen consumption between minute 2 and minute 3 differed by only 0.4% and 0.3% on average respectively ($P = 0.54$; $P = 0.63$). Table 2 shows the results of the submaximal and performance test variables during the study for each shoe condition.

Submaximal variables. For RE there were significant effects for shoe condition ($P < 0.001$; $\eta^2 = 0.572$), shoe–group interaction ($P = 0.001$; $\eta^2 = 0.123$), and shoe–velocity interaction ($P = 0.04$; $\eta^2 = 0.112$), but not for shoe–group–velocity interaction ($P = 0.14$). Running economy significantly improved with the increase of shoe LBS in the trained group at the slow (11.86 ± 0.93 vs 11.41 ± 0.93 W·kg⁻¹; $P < 0.05$) and fast velocities (16.39 ± 1.24 vs 15.89 ± 1.24 W·kg⁻¹; $P < 0.05$) and in national runners at the fast velocity (20.78 ± 1.18 vs 20.35 ± 1.45 W·kg⁻¹; $P < 0.05$), but was not significantly different in the national group at slow velocity (Fig. 2). Moreover, at 13 km·h⁻¹ a significant difference was found between the control and experimental shoe condition for the whole sample ($n = 27$) (15.86 ± 1.35 vs 15.50 ± 0.98 W·kg⁻¹; $P < 0.005$) and shoe–group interaction ($P = 0.031$; $\eta^2 = 0.051$). Correlation analysis showed that runners with poor RE (measured with control condition) improved their RE more when using the shoes with higher LBS than more economical runners ($r = 0.55$; $P = 0.04$).

Regarding the spatiotemporal parameters, we found significant effects for shoe condition ($P < 0.001$; $\eta^2 = 0.673$) and shoe–group interaction ($P = 0.010$; $\eta^2 = 0.241$), but not for shoe–velocity interaction ($P = 0.719$; $\eta^2 = 0.005$) and shoe–group–velocity interaction ($P = 0.895$; $\eta^2 = 0.001$) on step length. The step length increased in trained runners at slow velocity (0.92 ± 0.05 vs 0.93 ± 0.05 m for control and increased LBS; $P < 0.05$; ES = 0.629) and fast velocity (1.27 ± 0.073 vs 1.28 ± 0.07 m; $P < 0.05$; ES = 0.926), and in national runners at slow velocity (1.29 ± 0.06 vs 1.31 ± 0.06 m for control and increased LBS $P < 0.05$; ES = 1.343) and fast velocity (1.59 ± 0.08 vs 1.61 ± 0.08 m; $P < 0.05$; ES = 1.524), with larger increases in national runners at fast velocity (shoe–group interaction). Increased LBS condition showed a decrease in step frequency ($P < 0.001$; $\eta^2 = 0.539$), but without

TABLE 2. Results for variables analyzed during the study for each shoe condition.

	Trained			National			Shoe condition	Shoe x Group
	Control	Increased LBS	Cohen’s <i>d</i>	Control	Increased LBS	Cohen’s <i>d</i>		
Energy cost at slow velocity (W·kg ⁻¹)	11.86 $\pm$ 0.93	11.41 $\pm$ 0.93*	0.483	15.49 $\pm$ 0.95	15.36 $\pm$ 0.99	0.133	<0.001	0.020
Energy cost at fast velocity (W·kg ⁻¹)	16.39 $\pm$ 1.24	15.89 $\pm$ 1.24*	0.403	20.78 $\pm$ 1.18	20.35 $\pm$ 1.45*	0.32	<0.001	0.167
RER at slow velocity	0.858 $\pm$ 0.050	0.848 $\pm$ 0.052	0.332	0.789 $\pm$ 0.052	0.796 $\pm$ 0.050	0.174	0.744	0.220
RER at fast velocity	0.922 $\pm$ 0.052	0.917 $\pm$ 0.064	0.162	0.908 $\pm$ 0.050	0.918 $\pm$ 0.053	0.220	0.801	0.316
Δ lactate (mmol·L ⁻¹)	0.18 $\pm$ 0.65	0.51 $\pm$ 0.65	0.384	0.39 $\pm$ 0.59	0.39 $\pm$ 1.05	0.007	0.371	0.392
Step length at slow velocity (m)	0.921 $\pm$ 0.048	0.929 $\pm$ 0.052*	0.629	1.289 $\pm$ 0.062	1.310 $\pm$ 0.057*	1.343	<0.001	0.060
Step length at fast velocity (m)	1.267 $\pm$ 0.073	1.277 $\pm$ 0.073*	0.926	1.585 $\pm$ 0.083	1.606 $\pm$ 0.077*	1.524	<0.001	0.034
Step frequency at slow velocity (step/min)	162.5 $\pm$ 7.2	160.4 $\pm$ 6.7*	1.654	165.4 $\pm$ 7.9	164.4 $\pm$ 7.7	0.376	<0.001	0.143
Step frequency at fast velocity (step/min)	170.4 $\pm$ 7.4	169.4 $\pm$ 6.7	0.816	177.9 $\pm$ 11.2	175.9 $\pm$ 10.2*	1.063	<0.001	0.117
Contact time at slow velocity (s)	0.288 $\pm$ 0.016	0.290 $\pm$ 0.016	0.405	0.221 $\pm$ 0.034	0.230 $\pm$ 0.022	0.361	0.138	0.361
Contact time at fast velocity (s)	0.226 $\pm$ 0.008	0.227 $\pm$ 0.008*	0.672	0.190 $\pm$ 0.006	0.191 $\pm$ 0.006	0.440	0.009	0.724
Flight time at slow velocity (s)	0.793 $\pm$ 0.334	0.084 $\pm$ 0.028	0.316	0.139 $\pm$ 0.020	0.142 $\pm$ 0.019	0.559	0.075	0.630
Flight time at fast velocity (s)	0.127 $\pm$ 0.020	0.128 $\pm$ 0.020	0.267	0.148 $\pm$ 0.021	0.152 $\pm$ 0.019*	0.940	0.008	0.041
Vertical oscillation at slow velocity (cm)	7.46 $\pm$ 1.46	7.51 $\pm$ 1.49	0.128	8.94 $\pm$ 1.25	9.14 $\pm$ 1.24*	0.563	0.136	0.242
Vertical oscillation at fast velocity (cm)	8.22 $\pm$ 1.16	8.24 $\pm$ 1.17	0.108	8.35 $\pm$ 1.35	8.36 $\pm$ 1.60	0.016	0.966	0.945
3000 m TT test (s)	644 $\pm$ 61	639 $\pm$ 59*	1.084	574 $\pm$ 21	569 $\pm$ 21*	1.813	<0.001	0.917
HR during 3000 m (bpm)	177.91 $\pm$ 7.44	176.11 $\pm$ 7.06	0.248	177.33 $\pm$ 8.32	176.00 $\pm$ 9.62	0.147	0.124	0.592

Data are presented as mean $\pm$ SD.

* Different from conventional shoe at $P < 0.05$.

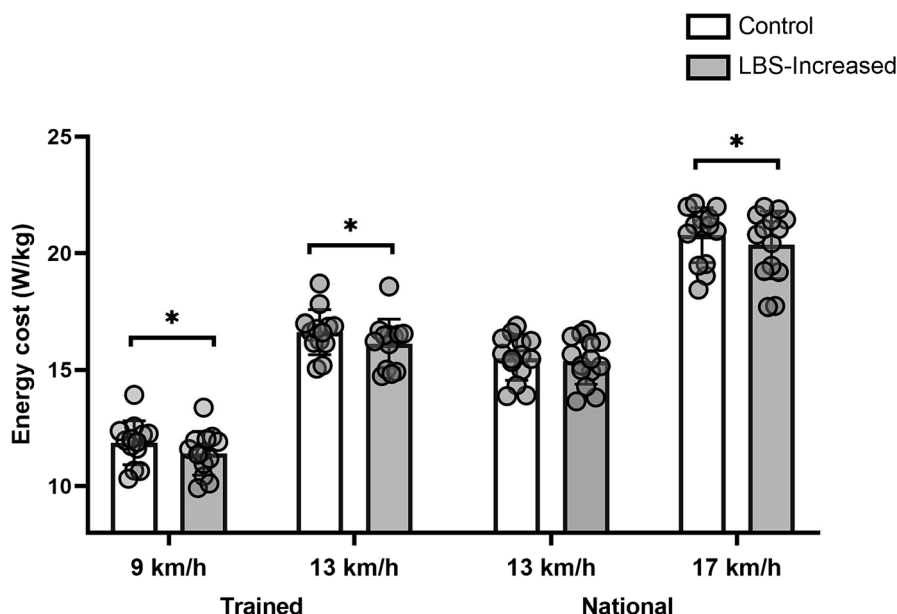


FIGURE 2—Running economy ($\text{W}\cdot\text{kg}^{-1}$) at $13\text{ km}\cdot\text{h}^{-1}$; Bar graphs represent mean values and circles represent runners. $*P \leq 0.05$ during *post hoc* comparisons when main effect of footwear significant. $n = 27$.

significant shoe-group ($P = 0.91$), shoe-velocity ($P = 0.833$) or shoe-group-velocity ($P = 0.175$) interaction effects. Step frequency decreased at slow velocity in trained runners (162.5 ± 7.2 vs 160.4 ± 6.7 steps/min for control and increased LBS; $P < 0.05$; ES = 1.654) and at fast velocity in national runners (177.9 ± 11.2 vs 175.9 ± 10.2 steps/min; $P < 0.05$; ES = 1.063). There was no significant main effect of shoe condition in contact time ($P = 0.081$; $\eta^2 = 0.121$) or shoe-velocity interaction ($P = 0.12$; $\eta^2 = 0.063$), but there were significant shoe-group ($P = 0.033$; $\eta^2 = 0.317$) and shoe-group-velocity ($P = 0.043$; $\eta^2 = 0.102$) interaction effects on contact time with an increase in contact time in trained runners at the fast velocity (0.226 ± 0.008 vs 0.227 ± 0.008 s for control and increased LBS; $P < 0.05$; ES = 0.672). There were significant effects of shoe condition in flight time ($P = 0.015$; $\eta^2 = 0.226$), and shoe-group interaction ($P = 0.045$; $\eta^2 = 0.145$), but there were no significant shoe-velocity ($P = 0.51$; $\eta^2 = 0.017$) or shoe-group-velocity ($P = 0.295$; $\eta^2 = 0.044$) interaction effects on flight time with a significant increase in flight time at fast velocity (0.148 ± 0.021 vs 0.152 ± 0.019 s for control and increased LBS; $P < 0.05$; ES = 0.940) in national runners.

For vertical oscillation there were no significant effects for shoe condition ($P = 0.351$; $\eta^2 = 0.035$), shoe $\times$ velocity interaction ($P = 0.14$; $\eta^2 = 0.024$) or shoe-group-velocity interaction effect ($P = 0.561$; $\eta^2 = 0.014$), but there was a significant shoe-group interaction ($P = 0.04$; $\eta^2 = 0.201$) with a significant increase in vertical oscillation at slow velocity (8.94 ± 1.25 vs 9.14 ± 1.24 cm for control and increased LBS; $P < 0.05$; ES = 0.563) in national runners.

Performance test. There was a significant main effect of shoe condition in the 3000 m TT ($P = 0.031$; $\eta^2 = 0.262$), but there was no significant shoe-group interaction ($P = 0.342$). The 3000 m TT test improved in trained ($10:44 \pm 1:01$ vs $10:39 \pm 0:59$ [mm:ss] for control and increased LBS; $P < 0.05$;

ES = 1.084) and national runners ($9:34 \pm 0:21$ vs $9:29 \pm 0:21$ [mm:ss]; $P < 0.05$; ES = 1.813). HR during 3000 m TT test was not different in trained runners (177.9 ± 7.4 vs 176.1 ± 7.1 bpm for control and increased LBS; $P > 0.05$; ES = 0.248) and national runners (177.3 ± 8.3 vs 176.0 ± 9.6 ; $P > 0.05$; ES = 0.147). Correlation analysis showed that individual RE improvement during submaximal treadmill running was not correlated with individual 3000 TT improvement ($r = 0.29$; $P = 0.19$).

Participants ran faster in the experimental shoe condition during lap 2, 4, 6 and 7 ($P < 0.05$) (Fig. 3). When comparing the velocity of the first 200 m with the velocity of each following lap, there were a significant decrease between the velocity of the first 200 m and the velocity of laps 1, 2, 3, 4, ($P < 0.05$), 5 and 6 ($P < 0.01$) in the control condition, whereas in the increased LBS condition there were only a significant decrease between the velocity of the first 200 m and that of laps 4, 5 and 6 ($P < 0.05$) (Fig. 3).

DISCUSSION

The aim of this study was to examine the influence of increased shoe LBS on RE, spatiotemporal parameters and distance performance in trained versus national runners at different running velocities. The main findings were a greater improvement in RE in the trained group (3.80% when running at slow and 3.24% when running at fast velocities) than in national runners (a nonsignificant $\sim 1\%$ and a significant 2.11% when running at slow and fast velocities, respectively), as well as improvement in 3000 m TT test and more constant pacing during the TT in both groups when wearing shoes with increased LBS compared with control shoes. Moreover, the improvement in RE was accompanied by spatiotemporal modifications, mainly an increase in step length.

Our RE results are in agreement with previous reviews and meta-analysis (12,21) where a positive association between

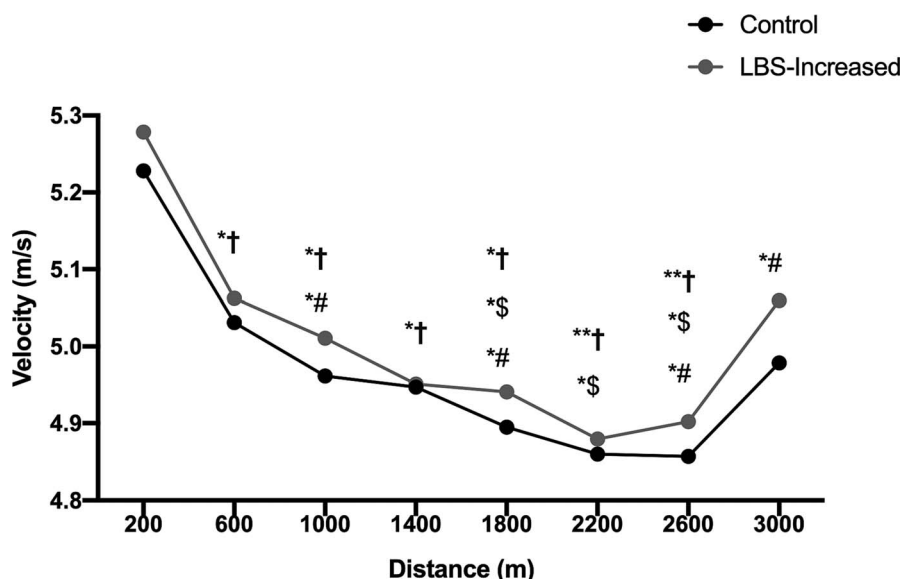


FIGURE 3—Running velocity in each lap for each condition. *Post hoc* analysis: #Different from Control and increased LBS; \$ Different from velocity of 200 m lap in increased LBS condition; †Different from velocity of 200 m lap in control condition; * $P < 0.05$; ** $P < 0.01$.

LBS and the metabolic cost of running was found. However, in the current study this association was not found in national runners at slow velocity ($13 \text{ km} \cdot \text{h}^{-1}$). To date, the exact mechanism by which shoes with increased LBS improve RE is unclear (9). Increased LBS reduces the dissipation of mechanical energy at the metatarsophalangeal joint by reducing its range of motion (9). Ankle joint mechanics could also be altered by an increase of its push-off moment due to a greater lever arm (9) which allows the triceps surae muscle fascicles to operate at a more favorable length reducing their metabolic demands (37). Moreover, the effect produced by the increased LBS and the rocker geometry of the midsole of AFT as used in our study might increase the force acting on the heel, which would result in increased propulsion (38) during running. In our study both shoe conditions had the same midsole geometry, so it appears that AFT shoes with increased LBS are particularly effective in improving RE when having a rocker geometry.

The interaction of the mechanisms explained above could explain the different results reported previously. Previous studies have reported an average improvement between 2.8% and 4.2% in RE, with a wide variability ranging from -8.6% to 13.3% in national runners (2–4) while running at absolute velocities from 14 to $18 \text{ km} \cdot \text{h}^{-1}$, and slightly lower improvements (less than 2%) in recreational/trained runners while running from 10 to $14 \text{ km} \cdot \text{h}^{-1}$ (5,18) in groups of a slightly lower level than the one used in our study. However, all the studies mentioned above, used an AFT shoe as the experimental condition and a different traditional shoe as the control condition. Our study evaluated trained and national runners in the same study comparing footwear conditions that were identical with only a difference in LBS. Our results show a greater improvement in trained runners (3.80% when running at slow and 3.24% when running at fast velocities) than in national runners (a nonsignificant $\sim 1\%$ and a significant 2.11% when running at slow and fast velocities, respectively). We also found differences between groups

(shoe–group interaction) and between velocities (shoe–velocity interaction) as the LBS of the shoes increased. As opposed to a recent study suggests that higher level runners benefit more than lower-level runners (19) when used shoes with increased LBS. Some studies have highlighted the notion of an individually optimal LBS to improve RE. McLeod et al. (16) suggest that LBS would depend on the runner's velocity and running pattern. Oh and Park (15) showed that the optimal LBS could be related to the runner's natural metatarsophalangeal joint flexion range, therefore differences in LBS of the experimental conditions between studies and relative velocity could explain the contradictory results between studies. For example, Chollet et al., (19) study, the increased LBS condition used did not improve RE on average over the whole sample analysis in comparison to control condition, whereas our experimental condition improved RE at $13 \text{ km} \cdot \text{h}^{-1}$ in the whole sample ($n = 27$), so our increased LBS condition may be more optimal for a larger number of runners (using similar increases in LBS in the experimental condition) running at this velocity.

Moreover, it seems to be that runners with worse RE are assisted in a greater extent than more economical runners ($r = 0.55$; $P < 0.05$). These results are in relation to a recent experimental study and meta-analysis (21) where higher level (elite) runners showed less RE improvement when using AFT than national runners. Authors suggested that the world-class cohort was already more economical when running in the traditional running shoes and therefore would not benefit as much when compared with the national runners population (21).

Currently, there is controversy as to whether the improvement in RE with shoes with increased LBS is greater at faster velocities (2,39). Day and Hahn (40) suggested that the metatarsophalangeal joint may increase its dorsiflexion velocity as the joint angle excursion increases with running velocity. In our study, we did not measure the influence on the metatarsophalangeal joint, but our results, indicate that there is no relationship between higher

absolute velocity and higher RE improvement with shoes with higher LBS as Hoogkamer et al. (2) suggested. We can assume that depending on the level of the runner, their metatarsophalangeal joint will adapt (increasing or reducing the joint stiffness) to their running velocity and the improvement will be similar for a similar relative velocity, i.e., the higher the relative velocity the more the RE will improve, but it will not help a national runner running at a higher absolute velocity any more than a trained runner if the relative velocity is similar. However, we have no data that could justify this and to our knowledge, we are not aware of any previous articles that could explain this. Therefore, more studies are needed on metatarsophalangeal joint function in different levels of runners running at different speeds.

The slight changes in RE found at submaximal intensities were accompanied by changes in spatiotemporal parameters such as an increase in step length in both groups and at all velocities evaluated and a decrease in step frequency at slow velocity in trained runners and at fast velocity in national runners. In addition, contact time increased in trained runners at the fast velocity and vertical oscillation increased at the slow velocity and flight time increased at the fast velocity in national runners. The biomechanical mechanisms of improving RE and/or running performance with LBS increased footwear have been investigated by many researchers (2–4,9,41). Previous studies have shown that the differences in ground reaction force and/or spatiotemporal parameters between increased LBS footwear and traditional shoes were relatively small, and probably not pronounced enough to explain the magnitude of the improvement in RE during running in this type of shoes (for review, see Ortega et al., (9)).

However, our recent meta-analysis suggests that the improvement in RE is accompanied by an increase in step length and contact time (12). In the current study, we found an average increase in step length of ~2 cm in national runners and an increase of ~1 cm in trained runner along with a proportional decrease in step frequency. Previous studies also reported a similar increasing step length (3,4,42) and step frequency decrease (41,42) in shoes with greater LBS compared with traditional shoes in national runners and trained runners (24). Our results show that trained runners, as opposed to national runners, increased contact time 0.01 s during running in shoes with increased LBS. The slightly increase of contact time could be related to the need to generate longer propulsion phase (13). By increasing the contact time the lower limb muscles have more time to contract in a stiff shoe and could therefore shorten at slower velocities compared with a control shoe, which is related to better RE (43). Moreover, although national runners increased step length more than the trained runner's group, the national runners group also increased flight time (interaction shoe–group and shoe–group–velocity) and vertical oscillation (only at slow velocity). It has been shown that across different runners greater vertical oscillation is related to worse RE (44), so increased vertical oscillation at slow velocity in the national runners could potentially explain this group's failure to improve the RE at slow velocity with shoes with increased LBS. Keeping in mind that the running pattern and running kinematics are a dynamic system with several intercon-

nected variables (45), the different energy savings between groups might be due to the combination of many but small biomechanical adaptations (42).

We used an outdoor 3000 m TT test to assess short distance running performance with both shoe conditions. Our results provide novel evidence that shoes with increased LBS elicit better short distance running performance than control shoes, similarly for runners of different levels (0.78 and 0.87% for trained and national runners, respectively) while running at intensities higher than those measured in the RE tests. Notably, the difference in the 3000 m TT test between experimental (increased LBS) and control was small (5 s in both groups) which is smaller than previous results for 5000 m and 3500 m test in trained runners in studies using AFT vs control (5,18). Hébert-Losier et al., (5) found a performance improvement ranged from 3.8% to 8.2% when trained runners ran in an AFT shoe compared with their own traditional footwear. Eleven of 18 trained runners performed best with the shoes with the highest LBS. In our study, all runners had their best TT result in the increased LBS condition.

When analyzing the effect of increased LBS on pacing during the 3000 m TT test, the runners had a smaller velocity difference between laps (i.e., more consistent pace) in the increased LBS condition, running faster than in the control condition. This could benefit the performance in middle- (e.g., 1500 m) and long-distance events (e.g., marathon) due to very little velocity changes across the whole race, which is important for performance (46,47).

To our knowledge, TT performance has not been quantified before for national/elite runners in this type of shoe. Retrospective studies on elite runners report an improvement of ~1% in events such as the marathon (6,7). In our TT tests the shoes with increased LBS assisted trained runners (0.78%) and national runners (0.87%) similarly (5 s). Also, individual RE improvement during submaximal treadmill running was not correlated to individual 3000 TT test improvement ($r = 0.29$; $P = 0.19$), similar to the study by Nielsen et al., (18). However, the overall relationship between improved RE (3.80–3.14% for slow and fast velocities) and improved running performance (0.78%) in trained runners in the present study agrees with RE results from Fuller et al. (48) and theoretical modeling of the relationship between RE and running velocity (49). For national runners the improvement in performance (0.87%) was higher than theoretically expected based on the improvement in RE (1–2%).

LIMITATIONS

Although the biomechanical parameters in treadmill running were similar to overground running, it must be considered that there are several differences, such as lower limb joint kinematics and contact time (50). This might have affected the comparison between submaximal intensities in the treadmill and the performance test in the athletics track. In addition, the 1% incline (31) could have affected in the running biomechanics of running in shoes with LBS increased at the submaximal intensities running, and since air resistance during overground running is proportional to velocity squared, using the same

treadmill incline to compensate for this at a range of different velocities is not any better than simply evaluating level treadmill running. Furthermore, the participants in this study were all male runners. Hence, it is possible that the present results are limited to male runners. Also, although the methodology for measuring RE was controlled (stabilization of the $\dot{V}O_2$ between the second and third minutes) the short duration of the trials could have affected the measurement of RE. Finally, the measurement errors from conducting only one trial (no balanced) per shoe condition and velocity could have affected our regression analyses with this data. Moreover, a test–retest variability of the RE and the 3000 m TT was not performed. So, the interday variability of the 3000 m TT could have affected even though the external conditions were controlled. And the RE trials were performed with on the same day and with the same calibration to avoid device variability error (51).

CONCLUSIONS AND PRACTICAL APPLICATIONS

To our knowledge, this is the first study to compare the effects of increased LBS between runners of different levels at different running velocities on RE and performance and with shoes conditions that were identical except for LBS. Contrary to our hypothesis, increasing the shoes' LBS improved RE to a greater extent in trained runners (3.80% when running at slow and 3.24% when running at fast velocities) than in national runners (a nonsignificant ~1% and a significant 2.11% when run-

ning at slow and fast velocities, respectively). However, this type of shoes improved short running performance (during 3000 TT test) similarly in both levels of runners. The improvements in RE were accompanied by small modifications in running kinematics that could explain the differences between the different levels of runners.

The results of this study compared an AFT and a control shoe with identical characteristics (except for LBS). Most major running shoe companies now have a new style of road racing shoe with a thick lightweight midsole and accompanied by a carbon fiber plate that increases the LBS of the shoe like the ones used in this study. Therefore, footwear design companies should consider the increase in LBS, depending on the runner's level and velocity when creating AFT.

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Authors contribution: V. R. C. conceived the research idea and design. F. G. M., S. R. B., J. J. S. helped with the research design and carried out data collection, data analysis, and drafting of the manuscript. J. M. G. R. and W. H. helped draft the manuscript. All authors have read and approved the final version of the manuscript and agree with the order of presentation of the authors.

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